

Reading the Data: An Analysis of Willamette University Library's E-Resource Usage

Emily Strickland, Sophia Rabbanian, Tiffany Truong
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Abstract

Willamette University's Library has a host of online journals from a handful of publishers. As Willamette changes and the library reevaluates its needs, the librarians are needing to consider which electronic materials to maintain, and which to let go. With no recent way to analyze the usage patterns of journals and articles, we performed analysis on usage data in tandem with enrollment data to make actionable suggestions on how to fairly maintain the available online resources. From our analysis, we have found that journal usage ebbs when school is not in session, and that the Spring and Fall terms generally have higher usage, with Fall having the most consistent patterns of high-usage. In the most recent school year, usage has decreased by 29%, possibly due to an increase in LLMs as a research tool.

Introduction and Motivation

Willamette University has 3 libraries: Mark O. Hatfield Library (Salem Main Campus), J.W. Long Law Library (Salem Law Campus), and the Albert Solheim Library (PNCA Portland Campus). Each library has their own physical resources as well as shares a collection of electronic resources among all libraries. This includes journals, research papers and articles, and often come in packages bought by the library from publishers. In recent years, the WU Librarians have had no avenue to analyze patterns in journal requests and usage.

The library staff desires to understand the Willamette community's online journal usage as a whole as well as how it relates to enrollment in different departments. Through our analysis, the librarians will be able to consider which electronic materials are the most important to maintain, and which to deprioritize. Since some departments are smaller, they may be more likely to be hit by cuts if their resources have lower usage request figures. Willamette's librarians are striving to optimize their resources while maintaining fairness between departments and being good stewards of research.

Research Questions

RQ1: What are the patterns in usage in regards to subject matter, departments, semesters and classes?

RQ2: Can we use the patterns in usage to make suggestions on which resources to prioritize, and which resources are not as important?

Impact & stakeholders:

The key stakeholders of this project include Willamette University's librarians, students, and faculty. Librarians need clear insights into which resources are worth investing in as both students, and the university, evolve over time. Students rely on the library for the reading and research materials that are essential to their coursework, so it is critical that relevant resources remain available and accessible. Faculty, in addition to also having access to the library's resources, benefit when the library's e-resources are tailored to student needs and usage patterns.

This allows faculty to confidently direct students to the library for materials that support student success in their courses.

If we are able to identify the usage patterns in electronic resources at Willamette's libraries, we will be able to support the librarians in making data-driven decisions to optimize and ensure that resources across different academic departments are tailored, relevant and accessible to students across disciplines.

Background & Related Work

Like many other institutions, libraries have navigated changes in usage and funding landscapes over the years. According to the Library Journal article, "Shifting Sands - Budget and Funding 2026" by April Witteveen, many libraries have moved towards supporting more e-content and digital materials as a result of growing interest and usage of digital materials[1]. At the same time, Witteveen notes that while increases in digital materials spending may appear as if it would lead to greater availability of digital materials, spending increases are likely a result of the rising costs of digital licensing from publishers.

Licensing costs of digital materials have led many public libraries to adjust their usage policies to manage rising costs of digital collections. For example, Multnomah County public libraries in Oregon have decreased the number of digital materials patrons are allowed to borrow despite growing interest in digital materials since 2025 [2]. According to Multnomah County libraries, this is due to several factors including:

- **Consumer vs Library Cost** - Each digital item costs more to purchase for a library than the consumer. For example, Multnomah County writes that an e-book copy of the 2024 Kristin Hannah title, *The Women* has a consumer cost of \$15, while a single library copy costs \$60 (**Image 1**).
- **Purchasing of Multiple Copies** - In addition to the high costs associated with library copies of digital materials, libraries are often purchasing multiple copies of each digital material to meet the demand of patrons adding to the costs of materials.
- **Licensing Terms** - Publishers allow libraries to lease materials based on time or total number of checkouts. Libraries must also renew each material to keep the materials in their library.

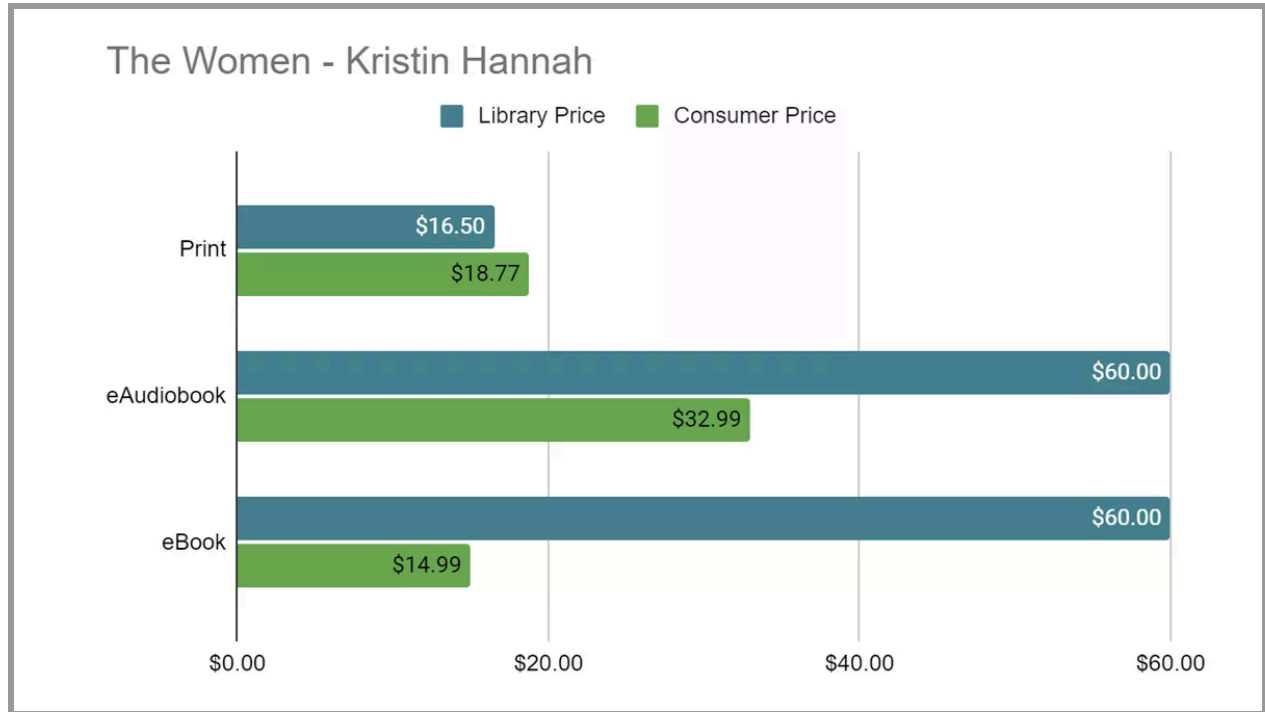


Image 1. Comparison between Library and Consumer costs of the book, The Women[2]. This bar graph shows the costs for Print, EAudiobook, eBook versions of Kristin Hannah’s The Women (2024) separated between Library Price and Consumer Price. The library price is often more expensive than the consumer price with digital materials such as EAudiobooks and eBook versions showing the largest disparity in costs between the consumer and libraries.

The rising costs of maintaining digital collections has led libraries like Multnomah County to reevaluate usage policies in order to maintain and optimize the usage of their digital collections. While Willamette University’s libraries are academic libraries with digital materials and research databases geared towards university students, they face similar questions and challenges to optimizing the cost and usage of digital collections. To understand and navigate these questions, data can provide insight into usage of different materials over time and student academic demographics to understand where there are gaps, or inefficiencies in the maintenance of Willamette’s digital materials.

Data as a Tool to Understand Library Usage

Utilizing data as a tool to understand library usage has long been highly advocated by libraries and library associations across the United States. In fact, the American Library Association (ALA) has established the Data-Driven Librarianship toolkit to capture key tools for libraries to support planning, improvement and advocacy. Tools include the ‘Impact Survey’ which allows libraries to survey patron’s technology needs, as well as the Public Library Data and Performance Measures to identify key metrics and performance resources to assist with decision-making [3]. Furthermore, the Institute of Museum and Library Services (IMLS) has

developed a dashboard using similar metrics developed by the ALA to provide librarians across the country a way to explore and compare circulation of materials and funding [4]. Regarding digital materials and research databases, the South Central Library System in Wisconsin, utilized similar e-resource datasets to create a dashboard that outlines the usage of the library's online resources over time [5]. Altogether these efforts indicate that there is a broader interest in utilizing data regarding usage of materials to guide decision making for libraries across the United States on both a federal and local level.

Data as a Tool to Understand Willamette's Libraries

While this project falls in a similar topic area as the previously mentioned public library data projects, it stands out as it integrates higher education contexts and enrollment data to understand a University's library usage. This provides the opportunity for our project to implement features from existing library data projects to understand the needs of a university's libraries. Furthermore, Willamette currently does not have an existing data pipeline to navigate and analyze library e-resource usage. This uniquely positions our project to develop a specified data pipeline for Willamette's libraries to make sense of usage data to increase efficiencies in the maintenance of the library's resources. The inclusion of these facets provides an opportunity for the project to streamline the available e-resources for students and have a clear impact on student education at Willamette University.

If we are successful in our analysis, our output will give the librarians data-driven insights on which resources are used the most and why. It will give them a better idea of which resources to keep or retire, while ensuring students are aware of the range of materials that are available to them. Similarly, this provides an opportunity to make progress on the development of a data pipeline to store, analyze and develop insights on library e-resource usage data that can be replicated for future school years. Ultimately, the findings and methods utilized in this project can support the development of data driven decisions for Willamette's libraries in the short term, while supporting the development of a framework for future analysis of higher education library data.

Data & Data Engineering

This project combines data from 2 different sources: WU Library Usage Request Reports and the Willamette University Factbooks. While the data was fairly clean and complete, there was some work to do for both data sources before it was possible to download it. We needed to label each journal with its subject, and write a script to get the enrollment information into a digestible file type.

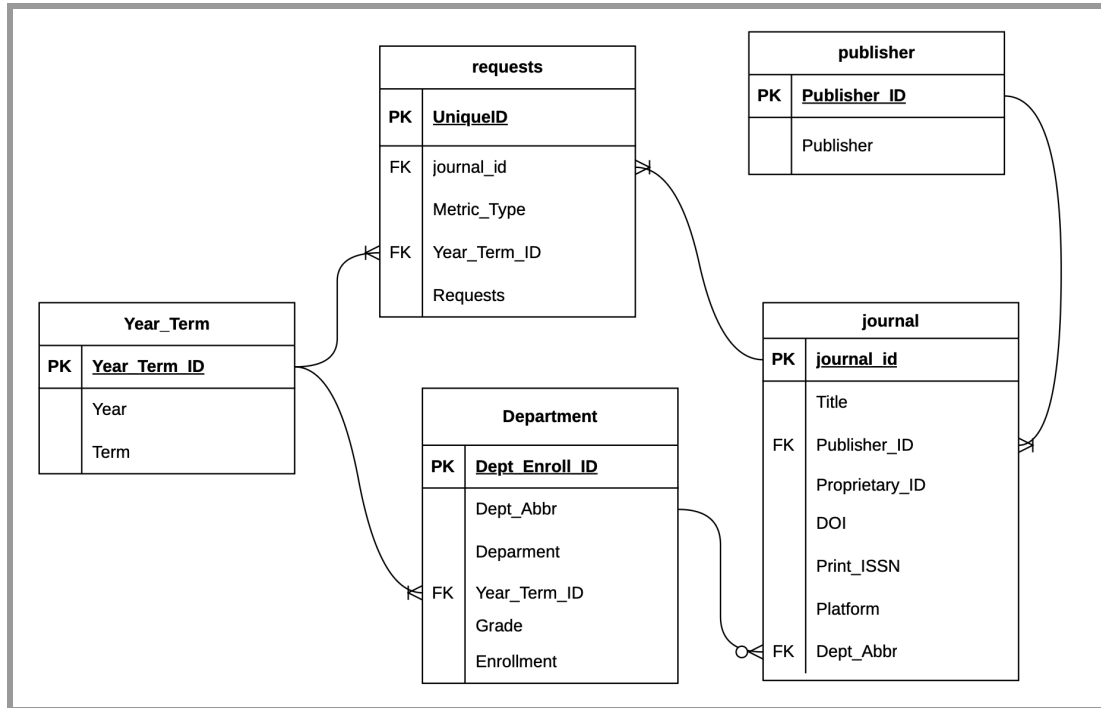
The Willamette Library usage requests data was supplied to us by a contact at the WU Library, and contained 7 different spreadsheets from 5 different publishers: Oxford University, Sage, Springer Nature, University of Chicago and Taylor & Francis. Each of these contained monthly data for the number of unique journal requests and the total number of journal requests.

The unique item request only counts how many unique journals or articles within each journal was requested. Ultimately, we used the unique requests metric for our analysis. This data spans from July 2023-March 2026. Since we wanted to analyze the relationship between enrollment in specific majors and the journal requests for journals in the same subject matter, we needed to classify each of the journals into their subject. This was a large endeavor since there were around 2,000 journals listed. Additionally, because there is some nuance as to what subject each journal belongs to, journals were labeled with their closest related subject based on our discretion, which was then checked for accuracy by a WU librarian. The data was then all compiled into one file, ensuring that all of the month columns lined up, as different publishers have data for slightly different time periods. Since we are curious about looking at enrollment and the enrollment data is only as granular as each term, we summed each month for the journal data to also only consider the term. To do this we needed to assign each month to a term, which we did as follows:

Term	Months
Fall	August-December
Spring	January-April
Summer	May-July

The enrollment data is publicly available on Willamette University's website and contains each semester's enrollment for every major broken down by sex and year in school. We chose the timespan to match the time for which we have journal requests data. This was from the Summer 2023-Fall 2025 (Spring 2026 data was not yet available at the time of our data ingestion). The data was stored as an .html on the website and we downloaded it as a .pdf. Before ingesting this data, we needed to write a script to read the data from the .pdfs to .csvs. We then combined all of this data into one file for ease in ingestion. After this was done, we were able to ingest the data and then wrangle. The important wrangling steps included summing the male and female data together as we did not care about sex in our analysis. Additionally, since we do not have enrollment data for the summer semesters, we had each summer adopt the previous spring's enrollment data, as we assume that very few (if any) students are declaring their major in May-July. Once these steps were completed, the data was ready to be transformed into the new schema.

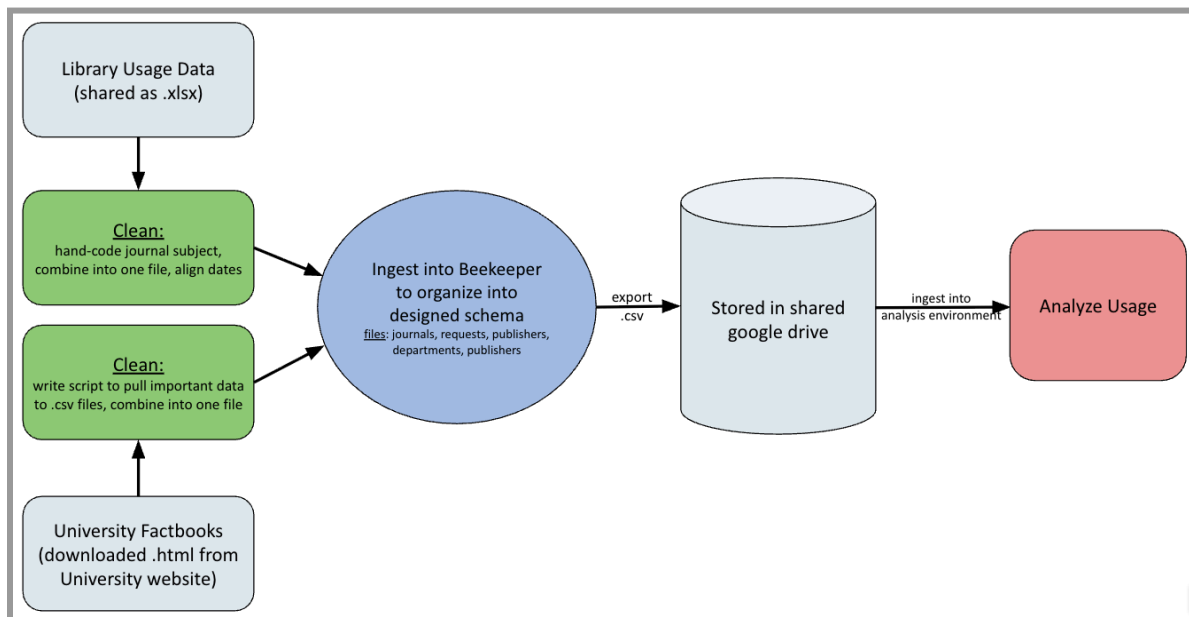
This data was downloaded and ingested into a PostgreSQL database where it was placed into staging tables. The new tables were then created, and the data was inserted from the staging tables into the new organized tables, which contained the appropriate foreign and primary keys to ensure ease in using joins for our analysis. Our organized schema can be seen below:



Once the data was in this format in our Postgres database, we exported each of the tables into .csv files so that we could download them into our analysis environments. These were stored in a shared google drive. This allowed us to easily perform our analysis asynchronously without having to re-wrangle the data every time we came back to our coding environment.

Pipeline

The data pipeline described above is visualized below:



Data Ethics

Our data raises few privacy concerns because none of it describes individual people. The journal usage reports are COUNTER TR_J1 reports, which record counts of article requests at the journal level. They contain no user information, no record of who accessed a resource, and nothing that could reidentify a student, faculty or staff member. Because there is no personally identifiable information in these files, there is nothing to redact and no consent process required.

The enrollment data comes from Willamette University's public Fact Books, which report aggregate counts of declared majors broken down by grade and sex. These are institution-level summaries, not student records, so they also contain no personally identifiable information. Both datasets are cleared for public use. As a result, no files in this project are access-restricted, and both the journal and enrollment data can live in our public repository.

Responsible Use:

The main responsible-use concern is fairness rather than privacy. Our analysis can influence which resources a department keeps or loses, so we are careful not to disadvantage smaller or underserved departments. We compare journals within each department rather than only across the whole collection, and we strive to make equitable suggestions backed by our findings. Because we do not want to cause harm to any department, we treat our prioritization output as suggestions for librarian review, not automatic decisions. Final decisions and actions rest with the Willamette librarians, who have strong domain knowledge in this subject matter and can best determine what resource allocation will best fit the needs of Willamette students and the broader Willamette community.

Methods & Evaluations

Approach

We use exploratory analysis and descriptive statistics rather than predictive modeling. The stakeholder question is descriptive: which resources are being used, by whom, and how usage is changing over time. A predictive model would answer a question the librarians did not ask, and with under three years of monthly data there is not enough history to train or validate one honestly.

Usage is measured with COUNTER Unique Item Requests rather than Total Item Requests. Total Item Requests counts repeated views of the same article, and different publisher platforms inflate that count differently depending on how their interface serves PDFs and HTML full text. Unique Item Requests counts each article once per session, which makes comparisons across publishers meaningful.

Journals are grouped into five publisher packages: Springer, SAGE, Taylor & Francis, Oxford, and University of Chicago. The library subscribes at the package level rather than by individual title, so package is the unit at which a keep or cancel decision is actually made. The raw files contain nine distinct platform values because some publishers report across multiple platforms. SpringerLink, Springer Nature Link, and nature.com are all Springer content and were collapsed into a single Springer package. A small number of records listed ResearchGate as the platform; these are not part of any library subscription and were dropped.

Requests are aggregated to the term level to match the granularity of the enrollment data, which is only published per term. For department-level comparison we normalize by enrollment, reporting unique requests per enrolled student rather than raw request counts, so that a large department does not automatically appear more engaged than a small one.

One part of RQ1 could not be answered. The COUNTER reports contain no course or class identifier, and neither does the enrollment data, so there is no way to link a journal request to a specific class. We report on subject, department, and term, and treat the class-level dimension as out of scope rather than approximating it.

Validation

There is no ground truth for what “correct” usage looks like, so validation focused on two things: whether the data supports the counts we report, and whether a finding survives reasonable alternative choices.

Data integrity. Oxford and Springer each supplied two export files covering overlapping time periods rather than one continuous file. Naively concatenating all seven spreadsheets double counts every journal and month that appears in both files, which inflates total requests by roughly 4.4%. We duplicate at the journal and term level, taking the maximum value where the two files disagree, before and aggregation.

Reporting floor. Publisher COUNTER reports omit titles that received no requests in a given period. A blank cell therefore means “not reported,” not “zero requests,” and the minimum observed value in the data is one request rather than zero. This means the collection cannot be described in terms of unused journals, and any month-over-month comparison across the full title list would confuse a change in reporting with a change in usage. For the time series we restrict to the 1,493 journals that appear in every month of the window, so the panel is constant and the trend reflects usage rather than coverage.

Coverage gaps. March 2026 appears in the data for University of Chicago only. Including it would show an apparent collapse in usage that is entirely an artifact of four publishers not having reported yet. That month is excluded from all time series, which sets the analysis window at July 2023 through February 2026.

Robustness. The year-over-year decline was checked separately within the publishers that supplied a single continuous file, so it is not an artifact of the deduplication step, and it appears in each of them rather than in one. Where a result depends on a chosen cutoff, we tested it under several cutoffs and report it only where the conclusion holds across them. An earlier version of the prioritization view used a consistency axis alongside volume, which we dropped after finding the two were almost perfectly rank correlated (Spearman rho of about 0.95) and were therefore measuring the same thing twice. The current version uses recency share rather than an arbitrary midpoint.

Results & Visualization

Figure 1: Journal Usage by Month, 2023-2026

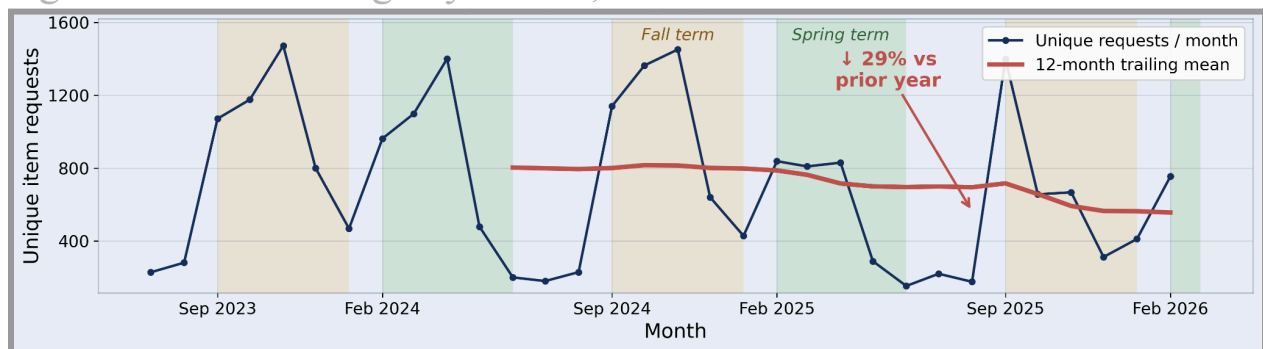


Figure 1. Journal usage by month, July 2023 to February 2026. Unique item requests per month, summed across the 1,493 journals reporting in every month of the window ($n = 1,493$). The red line is the 12 month trailing mean. March 2026 is excluded because only one of five publishers reported that month.

Figure 1 shows total unique item requests per month from July 2023 through February 2026, restricted to the 1,493 journals that reported in every month of the window ($n = 1,493$ journals). The vertical axis is unique item requests per month across the whole collection. Shaded bands mark Fall and Spring terms.

Usage tracks the academic calendar closely. Requests peak in the middle of each term, fall off sharply during winter and summer breaks, and reach their yearly low in June and July. Fall consistently produces the highest sustained usage of the three terms.

The clearer finding is the direction of the trend. The twelve month trailing mean fell from roughly 787 unique requests per month to roughly 556, a decline of 29% over the most recent year. Because the panel of journals is held constant, this is not a reporting gap. The decline also appears within the publishers that supplied a single continuous export file, which rules out the deduplication step as a cause. We report the decline as real and note that this analysis cannot identify its cause. The rise of large language models as a research tool is one plausible explanation, but nothing in this data tests it.

One feature of the series should not be read as a recovery. The isolated spike in September 2025 is driven by a single newly activated journal title rather than a broad return to earlier usage levels, and the series returns to trend the following month.

Figure 2: Departments Ranked by Usage Intensity per Student

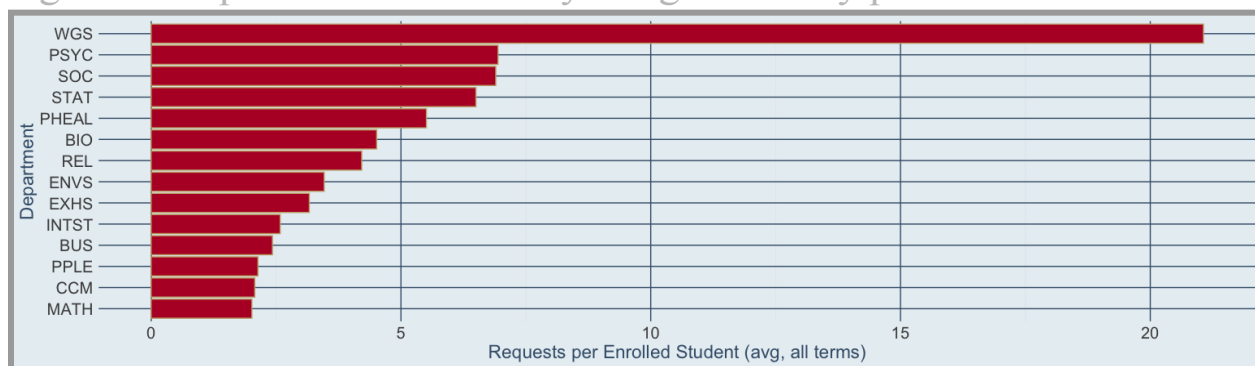


Figure 2. Journal requests per student broken down by department. Total unique item requests for each department divided by the average enrollment over all terms of available enrollment data.

Figure 2 shows the number of journal requests per student within each department, averaged across all terms present (Summer 2023-Fall 2025). The vertical axis displays the departments with the top 14 requests per student numbers. The horizontal axis denotes the number of requests per student.

Women’s & Gender Studies (WGS) is a huge outlier, with an average of 3.88 majors per term, but 957 total requests. While 957 is not a particularly high amount of requests (9th overall), having such low enrollment leads to the outlier. While there is no obvious reason for why this might be, we hypothesize that it could be a mix of (1) popular topics for research in WGS journals, which mostly focus on sexuality, gender identity, or women’s studies; (2) classes in other departments, like PNCA’s liberal arts, using WGS-labeled journals; or (3) use by other campus organizations like the Gender Resource & Advocacy Center (GRAC).

Psychology (PSYC) and Sociology (SOC) follow WGS with the next highest requests per enrolled student ratio with an average of 96.12 and 36.75 students per semester, and 5339 and 2028 total requests, respectively. These are much larger departments enrollment-wise, and also have a higher volume of journals, both ranking in the top 5 for journal count. Public Health (PHEAL), Biology (BIO), and Environmental Science (ENVS) should be mentioned here too, as they make up the rest of the top 5 departments by journal count, and also have consistently high enrollment.

Statistics (STAT) ranks fourth in requests per student, which supports the idea that statistical methods are used across many other disciplines. The same can be said for all other departments, so it is reasonable to assume that the requests were made by a mix of students from

different departments and other members of the Willamette community. Overall, this graphic shows that enrollment is not a strong indicator of journal usage, given the variation and the volume of out-of-major journal usage by students, faculty and staff alike.

Discussion, Recommendations & Limitations

Discussion & Recommendations

Analysis of usage patterns over the course of time indicates that while usage of e-resources often emulates the ebb and flow of semester terms; on a whole, usage of e-resources appears to be declining. While this report does not identify the cause, speaking with our stakeholders has suggested that shifting student research patterns including the rise of large language models, de-prioritization of journals as a resource and tool for research may contribute to this overall decline. While there are no definitive reasons that can be derived from our existing data and analysis, these usage patterns and discussions represent a larger need to understand shifting student research habits and classroom practices regarding what and how resources are discussed and promoted.

Similarly, variations in the number of requests for journals associated with different academic disciplines allow us to explore how journal topic areas can be a factor to consider when discussing which journal packages to invest in. As mentioned in the Results and Visualizations section, journals categorized under Women's & Gender Studies (WGS) were a significant outlier given the enrollment size of the WGS major. Our existing hypotheses for these sections include 1.) the popularity of the topic, 2.) cross disciplinary use of these journals across majors and schools, and finally, 3.) usage from other campus organizations such the Gender Resource & Advocacy Center (GRAC). While our findings indicate that enrollment based on academic disciplines is not a strong indicator of journal usage, the alignment of journal topic areas with overall student academic interests, organizational needs may be a necessary factor to consider. For example, if our hypotheses do hold true, Willamette's libraries stand to benefit from investing in journal packages that support journals of popular topics or topics that can be relevant to multiple disciplines or organizations on campus.

While these hypotheses require further research to fully understand and confirm their validity, this indicates an overall need to develop a better understanding of what student needs are regarding library resources. Understanding how students are navigating research, what research tools students are exposed to in classes, and finally, what resources/topics of interest are to the student body will allow Willamette's librarians to better tailor and optimize resources to the needs of Willamette's student body.

Limitations

This project analyzes the requests per student. However, the current e-resource usage data includes journal access from everyone in the Willamette community, not just students. This potentially inflates usage for each academic discipline. Similarly, our existing journal data does not reflect/delineate between journal requests submitted by out-of-major students and major students potentially impacting our findings. Finally, many journals don't fit neatly into one subject making it difficult to truly assess how usage differs across subjects/academic disciplines.

Future Work

Given our existing analysis and limitations of the data, there are several pathways for future work and research. Firstly, to gain more contextual understanding of the usage of e-resources at Willamette's libraries, we suggest doing further research into the student research habits and topics of interest to identify what resources would be of most interest to students. Additionally, because this project represents the first iteration of a data process to ingesting, cleaning and visualizing data from several sources of higher education data, we recommend continuing to refine this process. Refinement could include developing clearer/more concise categories of journals and building an accessible/easily replicable process for non-technical audiences to develop visualizations/analysis on a regular basis even after our team is no longer at Willamette University.

Conclusion

This project has presented an opportunity to examine how contextual factors such as student enrollment and topic area can impact the popularity and usage of e-resources. This presents a consequential research project given its timeliness during a time of changing library usage patterns and necessity to optimize library resources to the needs of university students. While our analysis indicates that overall usage of e-resources are trending towards a decrease and enrollment numbers are not an indicator of the level of usage of journals, our findings present an opportunity to further explore the underlying reasons for these patterns. Understanding how student research habits have changed or developed will support in understanding how to best shape resources to the current needs of students, and understanding what topics are of interest to the Willamette's students will provide more perspective on which journal topics and packages to invest in. Furthermore, through our ingestion and cleaning process, we have been able to develop an initial pipeline that can be further refined and adapted to the needs of Willamette's librarians to support future analysis of usage data to refine and optimize Willamette's library usage. In all, this project ultimately highlights the strengths, and opportunities of data as a tool to begin to optimize resources while maintaining fairness between departments and being good stewards of research.

Reproducibility

Repository. All code, the normalized data files, and the notebooks that produced the figures in this report are at <https://github.com/sophiarabb/data510-libraries>. Both the journal usage data and the enrollment data are cleared for public posting, so the repository is complete and requires no restricted files to run.

Data files. The analysis reads five normalized CSVs exported from our PostgreSQL database: requests.csv, journals.csv, departments.csv, and year_term.csv. These correspond to the schema shown in the Data and Data Engineering section.

Shared loader. A shared module, library_data.py, handles loading and the two transformations that must happen before any aggregation: deduplication of the overlapping Oxford and Springer export files, and collapsing the nine raw platform values into five publisher packages. All three team members load the data through this module so that the corrections are applied identically across our analyses.

Environment and dependencies. Python 3.13, pandas for analysis, pdfplumber for extracting enrollment tables from the Fact Book PDFs, and matplotlib for figures. PostgreSQL 16 for the database, accessed through Beekeeper Studio. No machine learning libraries are used, since the analysis is descriptive.

To reproduce. Clone the repository, install the dependencies listed in requirements.txt, and run the notebooks in the analysis folder in order. The notebooks read from the CSVs in the data folder and regenerate every figure in this report.

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